

ResidualGuard: Learning Collision Avoidance with Command Fidelity for High-Speed Wheeled-Legged Robots

Abstract—Active safe locomotion requires wheeled-legged robots to balance rapid collision avoidance with command fidelity at high speed. Existing risk metrics based on obstacle distance or time-to-collision often fail to distinguish true collision threats from safe pass-by motions, leading to unnecessary avoidance and command deviation. We propose ResidualGuard, which learns perception-aware velocity residuals above a fixed locomotion policy while limiting deviations from external velocity commands. ResidualGuard combines LiDAR-based local geometry estimation with locomotion-response encoding to provide environmental and motion context for residual control. Its core component, Ray-wise Dynamic Collision Risk (Ray-DCR), jointly considers robot footprint geometry and radial-tangential relative motion to evaluate the risk of candidate executed commands. During reinforcement learning, this risk signal guides the residual policy to retain nominal commands in low-risk situations and correct them under collision threats. Simulation results in static forest and dynamic corridor scenarios show higher collision-free completion and task progress with lower command deviation than existing avoidance methods. Fully onboard experiments on a Unitree Go2-W with a Livox MID-360 further demonstrate real-time collision avoidance at speeds up to 4 m/s.

I. INTRODUCTION

Learned wheeled-legged locomotion policies support high-speed motion through a velocity-tracking formulation [1]. This separates task specification from motion execution, providing a common interface for operators and upstream controllers. Unlike goal-reaching policies conditioned on a destination [2], our correction policy uses the external nominal velocity command as its task reference. It can therefore serve different command sources without requiring their goal representations, but must limit changes to the requested motion during avoidance. At high speed, following an obstacle-unaware command can rapidly lead to collision, whereas excessive corrections can redirect or halt the requested motion. We therefore study collision avoidance with limited command deviation while maintaining task progress. Figure 1 shows representative robot behaviors.

High-speed reactive avoidance must reconcile short reaction windows with the limitations of motion generation. Trajectory replanning requires time to update a plan and realize the resulting motion, which can become restrictive as obstacles approach rapidly [3]. Optimization-based safety filters modify nominal commands online, requiring reliable safety estimates and computation within the reaction window [4], [5]. Joint-level learned policies tightly couple avoidance and locomotion [2], whereas residual correction and command shielding build on existing controllers [6], [7], [8]. Yet command-level decisions can underuse locomotion agility when downstream response is treated as ideal [2].

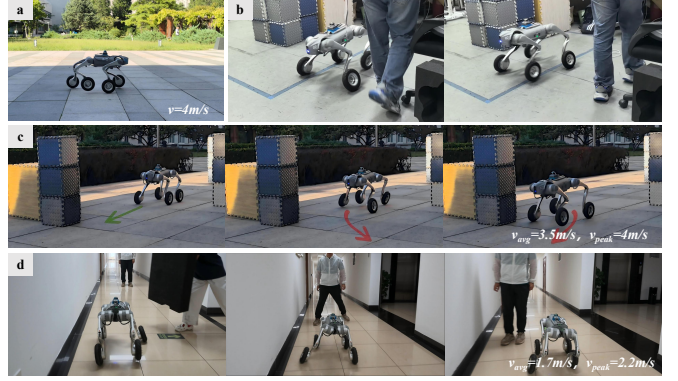


Fig. 1. High-speed locomotion and collision avoidance on the Unitree Go2-W. (a) The locomotion policy trained in this work reaches 4 m/s and serves as ResidualGuard’s fixed base controller. (b) Reactive obstacle avoidance from a stationary start. (c) Collision avoidance in an outdoor obstacle field at a peak speed of 4 m/s. (d) Avoidance of moving obstacles in an indoor corridor at a peak speed of 2.2 m/s.

The challenge is therefore to preserve the modular command interface while achieving prompt avoidance without unnecessary braking or steering away from the requested motion.

Learning such behavior requires distinguishing necessary avoidance from excessive intervention. Obstacle range can decrease during both a collision-threatening approach and a safe pass-by, so risk depends on footprint geometry and radial-tangential relative motion rather than range decrease alone [9]. Penalizing both similarly can induce overreaction, whereas relaxing the penalty weakens the learning signal for genuine threats. A useful correction must also account for execution: commanded velocity does not instantly become body motion, and recent locomotion history can provide downstream-response context [10], [11]. Meanwhile, rapid ego-motion and moving obstacles make LiDAR history span evolving geometry across different poses and times [12]. Learning should therefore connect candidate corrections to collision risk, while deployment requires response context and local geometry estimated with explicit motion and timing information.

We introduce ResidualGuard, a modular framework with a single residual correction policy between an external nominal velocity command and a fixed locomotion policy (Fig. 2). This policy is trained end to end through reinforcement learning (RL) to map the nominal command, local geometry, and locomotion-response context to a 3-DoF residual action. During RL, Ray-wise Dynamic Collision Risk (Ray-DCR) evaluates processed candidate commands using footprint-relative ray geometry and radial-tangential relative motion, providing a risk penalty that guides the timing, direction, and

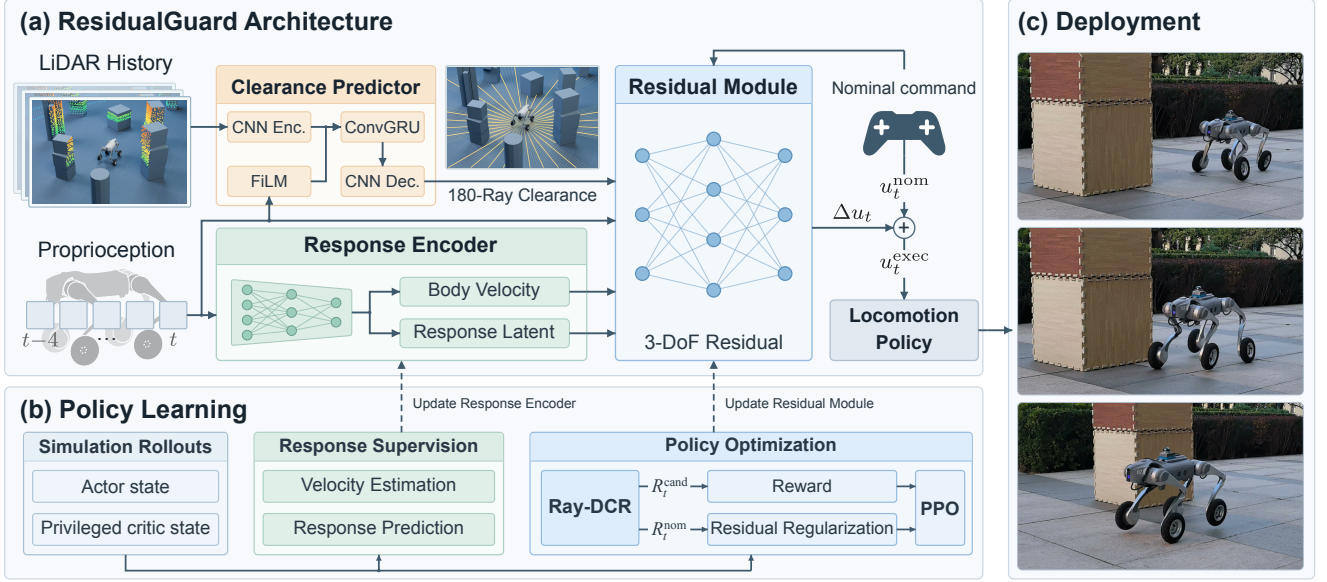


Fig. 2. ResidualGuard architecture and policy learning. (a) A single residual policy combines the nominal command, predicted radial geometry, and locomotion-response context to generate 3-DoF corrections above a fixed locomotion policy; $\Delta u_t = u_t^{\text{exec}} - u_t^{\text{nom}}$ denotes the effective correction. (b) Simulation rollouts support separate encoder supervision and PPO policy optimization. Velocity-estimation and response-prediction losses train the encoder; candidate Ray-DCR supplies a reward penalty, while nominal Ray-DCR gates command-fidelity rewards and residual regularization. (c) Fully onboard Go2-W deployment with a Livox MID-360; Ray-DCR is not evaluated at inference.

magnitude of corrections. Nominal risk additionally modulates residual regularization to favor small corrections in low-risk situations. A locomotion-response encoder supplies execution context, while a motion-conditioned clearance predictor estimates local radial geometry from LiDAR history, relative pose, and timing information. The same policy handles nominal-safe and threatening interactions without expert selection or policy switching; deployment requires neither online Ray-DCR evaluation nor safety-side optimization.

Our contributions are:

- 1) We propose ResidualGuard, a command-correction framework for high-speed wheeled-legged robots. Its residual policy combines motion-conditioned LiDAR clearance and locomotion-response context to learn 3-DoF corrections end to end, balancing collision avoidance and fidelity to external nominal velocity commands while retaining a fixed locomotion policy.
- 2) We introduce Ray-DCR, a candidate-command-dependent dynamic collision-risk signal for reinforcement learning. It combines footprint-relative ray geometry and radial-tangential relative motion to distinguish collision-threatening approaches from safe pass-by motion and guide command corrections, without online Ray-DCR evaluation.
- 3) We demonstrate higher collision-free completion and task progress with lower command deviation than the evaluated avoidance baselines in simulation, and fully onboard collision avoidance on a Unitree Go2-W among static and moving obstacles.

II. RELATED WORK

A. Reactive Avoidance Architectures

Residual shared autonomy learns small corrections to external actions without access to the user’s goal [7]. For legged robots, REASAN [8] places an absolute-command shield above a fixed locomotion policy and uses onboard LiDAR perception. ResidualGuard retains this modular command interface but learns residual corrections using locomotion-response context and radial geometry estimated with explicit motion and timing conditioning. We focus on the collision-avoidance-command-deviation trade-off during high-speed command execution, and compare against a Go2-W adaptation of REASAN under shared locomotion and command protocols.

Other reactive systems integrate avoidance more directly into motion generation. ABS [2] and REBot [3] coordinate agile or evasive locomotion with recovery policies, AWARE [13] selects locomotion experts, and Omni-Perception [14] learns LiDAR-conditioned joint actions for velocity tracking and avoidance. ResidualGuard instead trains a single intermediate residual correction policy end to end while retaining a fixed locomotion policy, without expert selection or policy switching. Unlike the motion-capture-based deployments reported for REBot and AWARE [3], [13], our system relies on onboard LiDAR for exteroception. It further demonstrates fully onboard obstacle avoidance at speeds up to 4 m/s, compared with the approximately 3.1 m/s peak speed reported by ABS [2].

B. Collision-Risk Learning and Runtime Filtering

Velocity obstacles model collision courses using relative motion [9], while control barrier functions constrain admis-

sible inputs through geometric safety conditions [5]. Barrier-inspired rewards [15] and learned safe-velocity regions [16] further incorporate geometric risk structure into policy learning. REASAN’s ray-wise TTC uses obstacle range and radial relative velocity [8]. In contrast, Ray-DCR evaluates processed candidate commands using footprint-relative ray geometry together with radial and tangential relative motion, explicitly accounting for pass-by motion when assessing candidate-command risk.

VOP-Nav [16] uses predicted safe-velocity regions as policy inputs and rewards. Runtime filters retain reachability assessment [17], adversarial predictive rollouts [18], or barrier-based intervention [19], [20]. ResidualGuard instead uses candidate risk for the RL reward and nominal risk for residual regularization, but executes learned corrections without online Ray-DCR evaluation or safety projection.

III. METHOD

A. Problem Formulation and Overview

We study collision avoidance with limited command deviation during high-speed wheeled-legged locomotion, with the aim of maintaining task progress. An operator or upstream module specifies the desired planar motion through a body-frame command $u_t^{\text{nom}} = [v_x^{\text{nom}}, v_y^{\text{nom}}, \omega_z^{\text{nom}}]^\top \in \mathcal{U}$. ResidualGuard adjusts this command before a fixed locomotion policy converts it into leg and wheel actions. The executed command and its deviation from the nominal command are

$$\begin{aligned} u_t^{\text{exec}} &= \text{clip}_{\mathcal{U}}(u_t^{\text{nom}} + \delta u_t^{\text{f}}), \\ \Delta u_t &= u_t^{\text{exec}} - u_t^{\text{nom}}, \end{aligned} \quad (1)$$

Here superscript f denotes the residual after smoothing, and $\mathcal{U}$ specifies component-wise command limits. ResidualGuard receives u_t^{nom} but not the upstream task goal.

We first pretrain a locomotion policy for high-speed velocity tracking and hold it fixed during residual-policy learning and method comparison. The residual correction policy is trained end to end through reinforcement learning with Ray-DCR risk signals, while the locomotion-response encoder is optimized with auxiliary velocity and response-prediction losses. The clearance predictor is trained independently to provide LiDAR-based radial geometry at deployment.

B. ResidualGuard Control Architecture

The residual policy combines the nominal command, radial geometry, and locomotion-response context to produce a 3-DoF correction through the interface in (1) (Fig. 2).

The nominal command alone does not specify the robot’s current motion. Prior locomotion methods infer latent context from recent state-action histories [10] and combine explicit velocity estimates with latent features [11]. Our locomotion-response encoder summarizes five frames of proprioception, previously executed commands, and locomotion actions in a causal history $H_t^{\text{loco}} \in \mathbb{R}^{5 \times 57}$. The encoder E_ψ , parameterized by ψ , maps the flattened history through a 128–64 MLP with ELU activations:

$$(\hat{v}_t^{\text{lin}}, z_t) = E_\psi(H_t^{\text{loco}}). \quad (2)$$

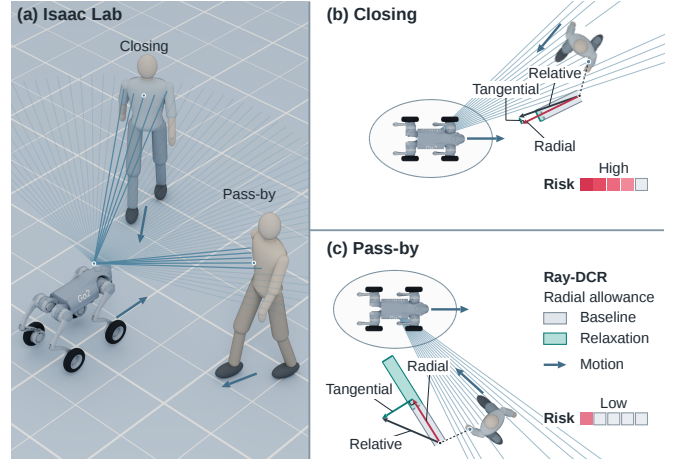


Fig. 3. Ray-DCR geometry for collision-threatening approaches and safe pass-by motion. (a) Example interactions in Isaac Lab. (b,c) Radial and tangential relative motion in closing and pass-by interactions, respectively. The gray and teal bars illustrate the distance-based allowance αg_i^{aff} and tangential allowance q_i , respectively, defined in the affine footprint geometry of (5)–(7). The added tangential allowance reduces the pass-by penalty. Sketches and risk indicators are schematic.

The encoder outputs a normalized body-frame linear-velocity estimate $\hat{v}_t^{\text{lin}}$ of $[v_x, v_y, v_z]$, including body- z velocity v_z , and an eight-dimensional L2-normalized response latent z_t . Both are available before the current correction.

We supervise the velocity head with current body linear velocity. An auxiliary response-prediction head uses z_t , the current planar response, and u_t^{exec} to predict the next normalized planar response $[v_x, v_y, \omega_z]$, where ω_z is yaw rate. This objective encourages the latent to retain history relevant to command execution. Auxiliary Smooth-L1 losses optimize the locomotion-response encoder and the training-only response-prediction head separately from the PPO policy objective. During PPO updates, the policy uses response context recorded during rollout.

The actor encodes the 180 normalized radial ranges described in Sec. III-D into a 64-D ray feature. This feature, $\hat{v}_t^{\text{lin}}$, and the 12-D vector of inertial observations, nominal command, and previous processed residual enter a single-layer, 256-unit LSTM. Its output is concatenated with z_t and passed through an MLP with hidden widths 512, 256, and 128 to a linear three-dimensional action-mean head. For a dimensionless action $a_t \in \mathbb{R}^3$, the processed residual is

$$\delta u_t^{\text{f}} = (1 - \beta)\delta u_{t-1}^{\text{f}} + \beta S_\delta a_t, \quad (3)$$

where S_δ converts actions to command units and $\beta \in (0, 1]$ is the current-sample weight of the exponential moving average. The nominal command is not smoothed, and S_δ does not impose a hard residual bound.

C. Ray-DCR for Reinforcement Learning

Ray-wise Dynamic Collision Risk (Ray-DCR) evaluates processed candidate commands through footprint-relative ray geometry and radial and tangential relative motion. Both closing and pass-by interactions can reduce obstacle range (Fig. 3); the tangential component distinguishes how an obstacle passes the footprint.

1) *Command-Dependent Motion Geometry*: At pre-action simulator state s_t , ray i has body-frame surface hit $p_i = d_i e_i$, with range d_i and unit direction e_i . The elliptical envelope expands with measured body velocities $v_{x,t}, v_{y,t}$:

$$\begin{aligned} a_t^E &= a_0 + \min(\tau_x |v_{x,t}|, m_{\max}), \\ b_t^E &= b_0 + \min(\tau_y |v_{y,t}|, m_{\max}), \end{aligned} \quad (4)$$

Here a_0, b_0 are longitudinal and lateral base semiaxes, τ_x, τ_y are speed-margin coefficients in seconds, and $m_{\max}$ caps the added margins. The speed-dependent expansion accounts for the increased spatial margin required at higher measured body velocities. Per-ray time and candidate arguments are suppressed. For a given pre-action state, the envelope is fixed across candidate commands within each risk evaluation and mapped to a reference circle of radius r_0 :

$$\begin{aligned} A_t &= \text{diag}(r_0/a_t^E, r_0/b_t^E), \\ r_i &= \|A_t p_i\|_2, \quad g_i^{\text{aff}} = r_i - r_0, \\ n_i &= \frac{A_t p_i}{\max(r_i, \epsilon_r)}, \quad t_i = J n_i, \end{aligned} \quad (5)$$

Here $J[x, y]^\top = [-y, x]^\top$, and $\epsilon_r > 0$ is a distance floor for normalization. The transformed range r_i gives the affine gap g_i^{aff} , measured relative to the mapped footprint boundary. The directions n_i and t_i define the corresponding radial and tangential axes in the transformed geometry. This affine normalization allows the same ray-wise risk construction to be applied around the anisotropic elliptical envelope.

For candidate command u_c , $\bar{v}_c \in \mathbb{R}^2$ and $\hat{\omega}_c \in \mathbb{R}$ denote its short-horizon translational-velocity and yaw-rate proxies, in m/s and rad/s. The hit obstacle's translational velocity $v_i^{\text{obs}} \in \mathbb{R}^2$ is expressed in the same body frame:

$$\begin{aligned} v_{\parallel,i} &= n_i^\top A_t (v_i^{\text{obs}} - \bar{v}_c - \hat{\omega}_c J p_i), \\ v_{\perp,i} &= t_i^\top A_t (v_i^{\text{obs}} - \bar{v}_c). \end{aligned} \quad (6)$$

Negative $v_{\parallel,i}$ denotes closing. The explicit yaw-point term $-\hat{\omega}_c J p_i$ contributes only to the radial component; the tangential component uses translational relative motion, preventing body-frame rotation alone from being interpreted as lateral escape.

2) *Ray-Wise Dynamic Risk*: The ray-wise margin B_i combines radial motion with gap and tangential allowances, all in m/s:

$$\begin{aligned} B_i &= v_{\parallel,i} + \alpha g_i^{\text{aff}} + q_i, \\ q_i &= \min \left(q_{\max}, k_\perp \frac{\ell_i^{\text{tan}}}{r_0} \frac{v_{\perp,i}^2}{\sqrt{v_{\parallel,i}^2 + v_{\perp,i}^2 + \epsilon_v^2}} \right), \end{aligned} \quad (7)$$

Here $\ell_i^{\text{tan}} = \sqrt{[r_i^2 - r_0^2]_+}$ is the tangent-length factor and $[x]_+ = \max(x, 0)$. Parameter α (s^{-1}) scales the gap, $k_\perp$ is a dimensionless tangential gain, and $q_{\max}, \epsilon_v$ (m/s) are the allowance cap and speed regularizer. Radial approach gives $q_i = 0$; tangential motion increases the allowance.

We pool dimensionless violations V_i , using velocity scale $s_B > 0$ and saturation cap $C > 0$. Invalid/no-return violations are zeroed before circular max pooling over ray

neighborhoods $\mathcal{N}_i$. Weights $w_i = \exp(\eta \tilde{V}_i) - 1$, with concentration parameter $\eta > 0$, emphasize larger pooled violations $\tilde{V}_i$:

$$\begin{aligned} V_i &= C \tanh \left(\frac{[-B_i/s_B]_+}{C} \right), \\ \tilde{V}_i &= \max_{j \in \mathcal{N}_i} V_j, \\ R(s_t, u_c) &= \frac{\sum_i w_i \tilde{V}_i}{C \sum_i w_i}. \end{aligned} \quad (8)$$

The dimensionless risk $R(s_t, u_c) \in [0, 1]$ is zero if all weights vanish.

3) *Risk-Guided Policy Learning*: Using the same pre-action state s_t , we evaluate the risk function for the processed command and the original nominal command: $R_t^{\text{cand}} = R(s_t, u_t^{\text{exec}})$ and $R_t^{\text{nom}} = R(s_t, u_t^{\text{nom}})$. Candidate Ray-DCR enters the RL reward with penalty weight $\lambda_R > 0$:

$$r_t = r_{\text{cmd},t} - \lambda_R \Delta t R_t^{\text{cand}} + r_{\text{fail},t}, \quad (9)$$

Here $r_{\text{cmd},t}$ sums tracking, nominal-direction retention, excess intervention, raw-residual first-difference, and command-limit reward rates, scaled by the control interval Δt . Tracking and excess intervention are modulated by nominal risk. The first three terms use u_t^{exec} ; smoothness uses the raw residual, and the limit term penalizes pre-clipping overflow. The terminal penalty $r_{\text{fail},t} = -\lambda_{\text{fail}} \mathbf{1}_{\mathcal{F}_t}$ uses weight $\lambda_{\text{fail}} > 0$ and indicator $\mathbf{1}_{\mathcal{F}_t}$ for a valid training hard-failure event, without Δt scaling.

Let θ denote the residual actor parameters and $\mu_{\theta,t}$ its action mean. We use nominal risk to gate an auxiliary actor loss, since a low-risk executed command may already contain a large correction:

$$\begin{aligned} L_{\text{reg}}(\theta) &= \mathbb{E}_{\text{valid}} \left[G_{\text{reg}}(\text{sg}(R_t^{\text{nom}})) \right. \\ &\quad \left. \times \frac{1}{3} \sum_{j=1}^3 \left(\sqrt{1 + ((S_\delta \mu_{\theta,t})_j / \epsilon_j)^2} - 1 \right) \right]. \end{aligned} \quad (10)$$

This auxiliary actor loss penalizes the raw mean $S_\delta \mu_{\theta,t}$ before residual processing and clipping; it does not enter the reward or return. Here $j = 1, 2, 3$ indexes the v_x, v_y, ω_z components, and $\epsilon_j > 0$ has matching units. The expectation averages valid nonterminal samples. The gate $G_{\text{reg}} \in [0, 1]$ decreases smoothly from one to zero as nominal risk increases; sg blocks gradients through R_t^{nom} .

PPO [21] trains the stochastic correction policy using returns from (9), while (10) contributes an additional actor-loss gradient. Simulator geometry, obstacle translation, and body-state truth provide risk signals, privileged critic inputs [22], and auxiliary targets. Nominal risk also conditions exploration. Deployment uses onboard observations and $a_t = \mu_{\theta,t}$, without Ray-DCR or the response-prediction head. Ray-DCR provides no formal safety guarantee.

D. Motion-Conditioned Clearance Estimation

REASAN predicts obstacle ranges from LiDAR history and penalizes overestimation [8]. Point-level compensation [23] corrects intra-grid ego-motion, but historical grids

retain different reference poses. Our predictor uses relative pose and observation age to interpret inter-grid changes.

Each range grid $G_k \in \mathbb{R}^{30 \times 180}$ is encoded by a circular CNN with channel widths 24, 48, and 64. The six-dimensional inertial feature I_k contains projected gravity and angular velocity, while the nine-dimensional M_k contains relative translation, relative rotation, observation age, pose-extrapolation duration, and validity, referenced to the latest grid. Their concatenation generates channel-wise scale and bias through FiLM [24]. A 64-channel ConvGRU [25], initialized to zero for each eight-frame window, fuses the conditioned features from oldest to newest. An upsampling decoder, elevation pooling, and a circular 1-D ray head produce 180 normalized radial distances. The predictor estimates the range target associated with the history window:

$$\hat{d}_{\tau_*} = d_{\max} P_{\varphi}(G_{1:K}, I_{1:K}, M_{1:K}), \quad K = 8. \quad (11)$$

Here φ denotes predictor parameters, and $k = 1, \dots, K$ indexes history frames from oldest to newest. The vector $\hat{d}_{\tau_*} \in \mathbb{R}^{180}$ contains predicted base-centered surface ranges in meters; τ_* is the simulator-label time, distinct from policy-use time. P_{φ} outputs ranges normalized by $d_{\max} = 4\text{m}$, without footprint subtraction. We call this range-based representation 180-ray radial clearance.

We train P_{φ} independently with

$$L_{\text{pred}}(\varphi) = L_{\text{metric}} + 0.20L_{\text{over}} + 0.25L_{\text{edge}} + 0.50L_{\text{temporal}}. \quad (12)$$

For target frame n and ray i , define range error $\Delta d_{n,i} = \hat{d}_{n,i} - d_{n,i}^{\text{GT}}$ (GT: ground truth). Let H be unit-threshold Smooth-L1, with error scale $h_d = 0.10\text{m}$ and overestimation margin $m_o = 0.10\text{m}$. Metric, edge, and temporal terms average $H(\Delta d_{n,i}/h_d)$, $H((\Delta d_{n,i} - \Delta d_{n,i-1})/h_d)$, and $H((\Delta d_{n+1,i} - \Delta d_{n,i})/h_d)$, respectively. Use circular ray indices and within-episode temporal pairs. L_{over} equally

averages global and critical-region means of $([\Delta d_{n,i} - m_o]_+/h_d)^2$. The critical region uses GT clearance $\leq 0.20\text{m}$ to a fixed ellipse with semiaxes $0.52/0.38\text{m}$. Empty critical regions use the global mean; no valid temporal pairs gives $L_{\text{temporal}} = 0$.

IV. EXPERIMENTS

We evaluate whether ResidualGuard avoids collisions while preserving nominal commands and task progress, then examine when it intervenes and how risk supervision, response context, and perception support this behavior.

A. Experimental Setup

We compare Nominal, REASAN-Filter (Go2-W adaptation) [8], Residual-TTC (Res-TTC), and ResidualGuard. Nominal directly executes the external command; REASAN-Filter outputs an absolute command. Res-TTC is trained independently with time-to-collision (TTC) supervision and shares ResidualGuard’s residual command-processing interface. All four share a frozen locomotion policy, robot dynamics, command limits, obstacle-blind nominal generator, and evaluation scenarios.

Static Forest uses staggered 8-, 12-, or 15-pillar layouts, uniform position perturbations, and mirrors. Dynamic Corridor has three sequential moving-cylinder interactions with sampled approach directions and speeds; nominal motion and two TTC levels determine initial positions and activation times. Both benchmarks use nominal speeds up to 4m/s , with 480 episodes across the four methods. Table I reports condition-balanced mean $\pm$ SD across the three paired scenario groups.

Success Rate (SR) is the fraction of valid episodes completed without collision. Any robot–obstacle or wall contact, fall, boundary violation, timeout, or in-task control abort counts as failure.

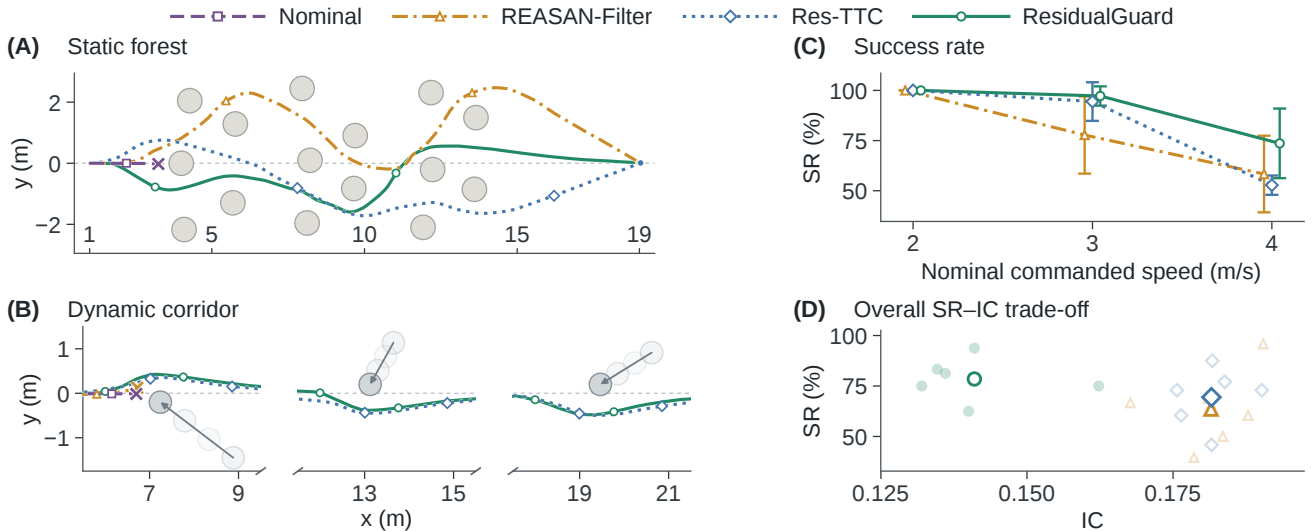


Fig. 4. Simulation comparison in Static Forest and Dynamic Corridor. (A,B) Recorded robot trajectories at a nominal speed of 3m/s ; arrows in (B) indicate obstacle motion. (C) Success rate (SR) versus nominal speed, with SD error bars. (D) SR versus intervention cost (IC): faded markers show six scenario aggregates per method, and larger outlined markers show aggregate means. Panels (C,D) compare the three avoidance methods.

TABLE I
SIMULATION PERFORMANCE IN STATIC FOREST AND DYNAMIC CORRIDOR

Method	Static Forest			Dynamic Corridor		
	SR (%) $\uparrow$	IC $\downarrow$	TP (%) $\uparrow$	SR (%) $\uparrow$	IC $\downarrow$	TP (%) $\uparrow$
Nominal	22.22 $\pm$ 19.25	0.000 $\pm$ 0.000	36.99 $\pm$ 19.70	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	24.64 $\pm$ 0.05
REASAN-Filter	87.50 $\pm$ 4.17	0.214 $\pm$ 0.039	93.57 $\pm$ 1.99	43.75 $\pm$ 6.25	0.126 $\pm$ 0.007	63.90 $\pm$ 7.17
Res-TTC	81.94 $\pm$ 8.67	0.202 $\pm$ 0.044	89.37 $\pm$ 5.17	62.50 $\pm$ 16.54	0.134 $\pm$ 0.002	79.50 $\pm$ 5.50
ResidualGuard	95.83 $\pm$ 4.17	0.177 $\pm$ 0.044	98.34 $\pm$ 2.31	70.83 $\pm$ 9.55	0.070 $\pm$ 0.007	82.79 $\pm$ 4.42

Intervention Cost (IC) averages the deviation in (1) over recorded control intervals:

$$D = \text{diag}(4.0, 1.7, 2.5),$$

$$IC = \frac{\sum_k \Delta t_k \|D^{-1} \Delta u_k\|_2}{\sum_k \Delta t_k}. \quad (13)$$

Here Δt_k is the duration of recorded interval k . The fixed scales in D match the maximum admissible magnitudes of v_x, v_y, ω_z in the common command set $\mathcal{U}$, in m/s, m/s, and rad/s; IC is dimensionless. Terminal Progress (TP) is $\text{clip}((x_T - x_0)/(x_{\text{exit}} - x_0), 0, 1)$, reported as a percentage, where T indexes termination, and $x_0, x_T, x_{\text{exit}}$ denote longitudinal start, terminal, and exit positions. IC and TP include successes and failures through first contact or other termination.

B. Simulation Evaluation

1) *Overall Performance*: Across both benchmarks, ResidualGuard combines higher collision-free completion and task progress with lower nominal-command deviation than both avoidance references (Table I). For example, in Dynamic Corridor its mean IC is 0.070 versus 0.134 for Res-TTC. These joint improvements indicate a better safety-intervention trade-off under the tested conditions.

Nominal has zero IC but lower SR and TP. Figure 4 complements the aggregate comparison with recorded trajectories and results across commanded speeds. The aggregate SR-IC summary in Fig. 4(D) uses predefined weights across the evaluation scenario sets rather than equal weighting. Corridor completion remains imperfect; the following analysis examines whether corrections are concentrated in interactions where nominal continuation would lead to contact.

2) *Behavior Analysis*: To examine when corrections are needed, we use a separate Threat-Pass-by study with predicted clearance (Fig. 5). A robot commanded at 3 m/s encounters an oncoming cylinder of radius 0.25 m moving at 1.5 m/s, across seven lateral offsets with perturbed mirror pairs. Alongside Res-TTC and ResidualGuard, we test w/o tangential, which removes tangential allowance from candidate and nominal Ray-DCR. Nominal continuation labels 11 collision cases as Threat and 17 contact-free completions as nominal-safe, independently of Ray-DCR.

All three avoidance methods complete the 11 tested Threat cases and the 17 nominal-safe cases. Yet their corrections differ when nominal continuation is already safe: ResidualGuard has mean IC 0.00534, compared with 0.0656 for Res-TTC and 0.0470 without tangential allowance. This ordering

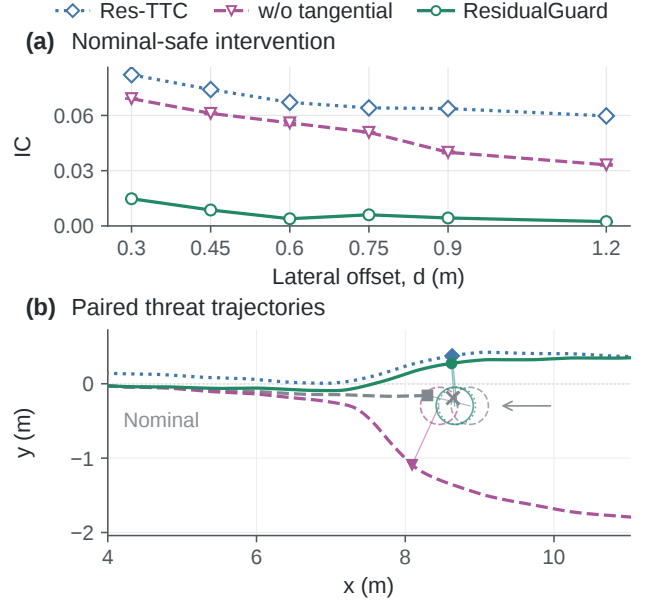


Fig. 5. Command intervention in nominal-safe and threatening interactions. Labels are determined by nominal continuation, independently of Ray-DCR. (a) Mean intervention cost (IC) versus lateral offset in nominal-safe scenarios, with paired 95% bootstrap confidence intervals. The displayed offsets contain $n = 1, 2, 2, 4, 4, 4$ nominal-safe scenarios from left to right. (b) Robot trajectories in a shared threatening interaction. Each robot marker is connected to the obstacle position at the corresponding time in that method's rollout.

persists across the displayed offsets in Fig. 5(a), with paired bootstrap intervals. Thus, ResidualGuard stays closer to the nominal command when intervention is unnecessary while preserving Threat completion in this study.

In the paired trajectory (Fig. 5(b)), ResidualGuard remains closer to the nominal centerline, while the no-tangential variant makes a larger lateral excursion. This difference is consistent with tangential relative motion in Ray-DCR supervision helping preserve the nominal motion direction during avoidance.

3) *Policy Ablations*: We evaluate tangential risk reasoning and response context at 4 m/s in the 15-pillar Forest and difficult Corridor, with 30 episodes per variant; IC and TP are reported as episode-level mean $\pm$ SD (Table II). Contact friction varies across scenarios and is shared within each scenario. The w/o response latent variant zeros the actor's eight-dimensional latent input while retaining the body-velocity head.

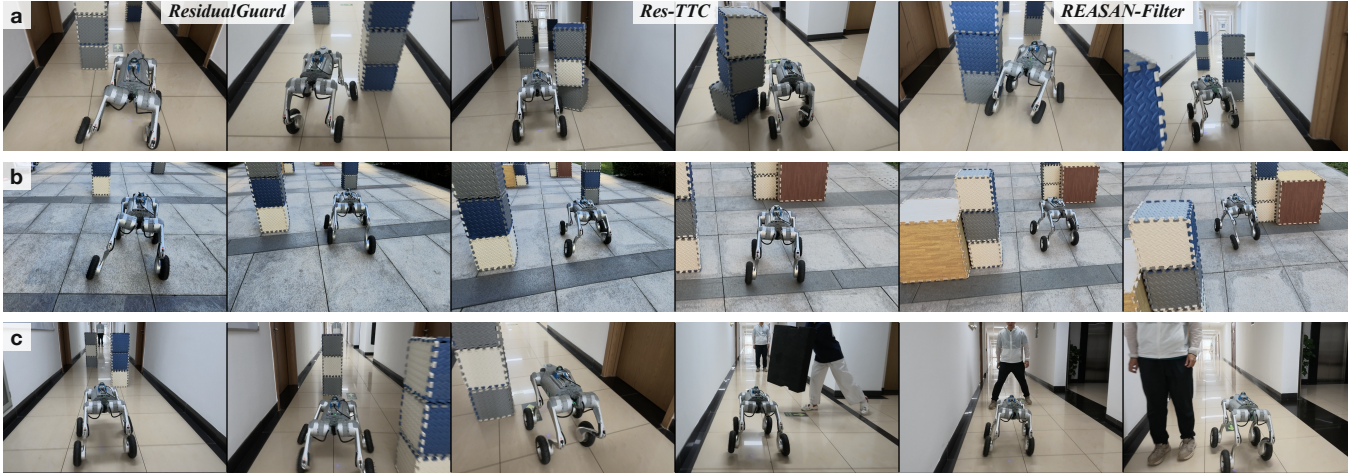


Fig. 6. Fully onboard evaluation on the Unitree Go2-W. (a) Three-method comparison in a 2 m-wide corridor with static pillars and a slippery tiled floor, at a peak robot speed of 1.8 m/s. The columns show ResidualGuard, Res-TTC, and REASAN-Filter from left to right; quantitative collision counts are reported in Table IV. (b,c) Representative ResidualGuard deployments in Outdoor Static Forest and Dynamic Corridor at robot speeds up to 4 and 2.2 m/s, respectively. Frames are ordered temporally within each sequence. All tests use LIO-based goal-to-command generation without live joystick steering.

TABLE II

POLICY ABLATIONS AT A NOMINAL SPEED OF 4 M/S

Variant	SR (%) $\uparrow$	IC $\downarrow$	TP (%) $\uparrow$
w/o tangential	66.7	0.378 ± 0.138	84.4 ± 23.3
w/o response latent	53.3	0.323 ± 0.149	70.5 ± 37.1
ResidualGuard	70.0	0.311 ± 0.164	90.0 ± 18.7

Without tangential allowance, mean IC rises from 0.311 to 0.378, while successes differ by only one trial: 21/30 for the full policy versus 20/30. In these stress conditions, tangential reasoning helps limit correction magnitude while retaining task completion, consistent with the behavior analysis above.

Without the response latent, successes fall from 21/30 to 16/30 and mean TP from 90.0% to 70.5%. With velocity estimation retained, this comparison supports the use of response history to maintain task execution under the tested friction variations.

4) *Perception Analysis*: We test whether motion conditioning improves the near-obstacle geometry supplied to the policy, using held-out episodes disjoint from training and development. Table III reports episode-equal mean $\pm$ SD over the 949 episodes containing critical samples. The two eight-frame, 4 m models share an architecture; w/o motion conditioning zeros explicit relative pose and timing inputs while retaining IMU, validity, and point-level compensation. REASAN Predictor [8], with 15 frames and a 3 m range, provides an additional reference beyond this conditioning ablation.

Critical samples have finite nonnegative ground-truth ranges and radial clearance of at most 0.20 m to a fixed evaluation ellipse with semiaxes 0.52/0.38 m. Ground truth alone defines this region, which differs from the speed-adaptive risk envelope. Critical-region MAE measures absolute radial-distance error; Dangerous Overestimation Rate (DOR) is the fraction of critical samples with predicted-minus-ground-truth range strictly above 0.10 m.

Motion conditioning reduces mean Critical MAE from

TABLE III

HELD-OUT CRITICAL-REGION CLEARANCE PREDICTION

Predictor	Critical MAE (m) $\downarrow$	DOR (%) $\downarrow$
REASAN Predictor	0.174 ± 0.094	18.6 ± 17.7
Ours w/o motion cond.	0.094 ± 0.085	19.3 ± 16.9
Ours	0.059 ± 0.065	7.7 ± 10.6

0.094 to 0.059 m and DOR from 19.3% to 7.7% (Table III). The unconditioned model has lower MAE than REASAN Predictor but a higher DOR. These results show that motion conditioning improves near-obstacle range accuracy and reduces dangerous overestimation in held-out perception tests.

C. Real-World Evaluation

1) *Hardware Setup*: ResidualGuard runs fully onboard a Unitree Go2-W with a Livox MID-360, using ROS 2, TensorRT/CUDA, and Unitree SDK2. FAST-LIO2 [23] supplies LiDAR-inertial odometry (LIO) for pose estimation and the perception front-end.

All real-world tests use a shared LIO-based goal-to-command rule without live joystick steering. A preset goal and the estimated pose determine the nominal command; only the command, not the goal, is supplied to the residual policy.

2) *Static-Corridor Comparison*: We compare REASAN-Filter (Go2-W adaptation) [8], Res-TTC, and ResidualGuard in a 2 m-wide indoor corridor with static pillars and a slippery tiled floor, at a peak robot speed of 1.8 m/s. Each method is evaluated in 10 trials with the same layout and goal-to-command rule, shared perception front-end, fixed locomotion policy, and command limits. A trial is counted once as a collision trial if any part of the robot contacts a pillar or corridor boundary. ResidualGuard records 0/10 collision trials, compared with 3/10 for REASAN-Filter and 5/10 for Res-TTC (Table IV).

3) *High-Speed and Dynamic Deployment*: We further deploy the complete ResidualGuard system in two settings.

TABLE IV

REAL-WORLD STATIC-CORRIDOR COMPARISON	
Method	Collision Trials ↓
REASAN-Filter	3/10
Res-TTC	5/10
ResidualGuard	0/10

In Outdoor Static Forest, the robot avoids successive tall pillars at speeds up to 4 m/s (Fig. 6(b)). In Dynamic Corridor, the robot reaches speeds up to 2.2 m/s while moving obstacles enter its path from either side; it avoids these intrusions and continues toward the goal (Fig. 6(c)). These deployments illustrate onboard collision avoidance among static and moving obstacles.

V. CONCLUSION

We presented ResidualGuard for high-speed collision avoidance with limited command deviation above a fixed locomotion policy. Ray-DCR-guided learning, locomotion-response context, and motion-conditioned clearance support corrections to external velocity commands. Simulations show higher collision-free completion and task progress with lower command deviation than the evaluated baselines, while fully onboard Go2-W deployment demonstrates obstacle avoidance at speeds up to 4 m/s.

The present system compresses surrounding geometry into a planar 180-ray representation. This representation discards vertical structure and is insufficient for height-dependent collision reasoning, such as overhangs, under-table passages, or obstacles whose traversability depends on height. Ray-DCR also relies on finite-range observations and a short-horizon approximation of relative motion, and serves as a training risk surrogate rather than a formal safety guarantee. Extending ResidualGuard toward height-aware or volumetric collision representations, richer dynamic-motion reasoning, and uncertainty-aware risk estimation would broaden its application to more complex three-dimensional environments while retaining the modular velocity-command interface.

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